COM748 Masters Research Project

Project Proposal Form

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Project Title
A Property Portal For Direct Landlord listing

Problem Statement (100 words)

In the UK, most rentals are completed through estate agents, plus listings advertised via commercial property websites such as Rightmove and Zoopla. Using these commercial sites requires landlords to use the services of an intermediary (usually an estate agent) which incurs costs, on average, of between £500 - £1,500 per tenancy. Small landlords alone make up a large share of the UK's rental housing supply and will continue to be impacted by the high transactional cost of working through an intermediary when renting out their property to tenants. The few existing direct-landlord platforms do not often provide the same level of functionality, usability, and trust features typically associated with agent-centric systems. The DirectLet Project will create a new, modern property listing site allowing landlord and tenant-to-interact directly by way of property listings, messaging, scheduling viewing appointments, processing payments, and verifying information, therefore helping landlords and tenants reduce costs and have an efficient rental-management experience.

Background Research (500 words)

The emergence of digital transformation has allowed for the elimination of many intermediaries in various sectors due to decreased necessity for physical intermediaries between service providers and their customers, and the facilitation of direct communication between service providers and customers. A prime example of this phenomenon can be found within the UK's property market. Historically, estate agents have served as the intermediary between landlords and tenants. Now, however, the growth of digital platform development has created new ways for landlords to directly promote and manage their properties. In their study, Jones and Turner [1] researched digital disintermediation as it applies to the UK property market and found that the use of digital platforms can lead to reduced transaction costs and increased effectiveness, however, there are still concerns about trust and user adoption of these platforms.

Well-designed properties for rent have an enormous effect on how well a rental property company will do in the marketplace. Chen, Wang, and Zhang [2], who focused on two-sided business design for real-estate type businesses, demonstrated the importance of balancing the funding and operations of the systems that facilitate rental property transactions between the tenant and landlord, in order to better serve both types of participants. In addition, they showed that resources with strong network effects, good channels of communication, and features that are centred on enhancing the user experience will increase the likelihood of participation by either or both types of participants in the rental property marketplace.

User Adoption of P2P Rental Systems is Influenced by Trust and Transparency. According to Yamamoto and Suzuki [4], trust, convenience, transparency, and ease of use are the major success factors affecting property rental platforms. They also indicated that if a verification system and communication tool were implemented to create confidence between users of the platform, the number of users participating in that platform would increase.

The use of new technology in the rental sector has been explored more recently through research. Rahman et al. [3] proposed a smart contract blockchain property rental system for the purpose of improving security and transparency in rental transactions. Their proposed approach enhances management of contracts, but primarily addresses security of the transactions themselves, rather than addressing other aspects of the entirety of the rental lifecycle (e.g., property discovery, communicating with landlords, scheduling, monitoring landlord performance).

While studies already exist related to disintermediation, platform economics, trust mechanisms, and new technologies, there is limited research on the integrated direct-lender rental platform that covers the full spectrum of the rental process. The majority of available solutions are generally limited in their capability to perform only stand-alone functions and do not offer an integrated solution that includes features for managing an entire property listing from beginning to end, including: Communication, Viewing Scheduling, Payment Processing, Analytic Reports, and Verification.

The project's goal is to create a full-featured platform (DirectLet) for landlords and tenants to transact directly with each other in a safe and secure environment. On DirectLet, landlords can list properties for rent, tenants can search for them with customized filtering options, communicate with landlords via messaging, schedule property viewings, pay rent online using credit cards and/or other electronic payment methods, view landlord analytics, and verify the identity and creditworthiness of all parties before engaging in any transacting activities. The research will aid further studies on digital disintermediation and two-sided marketplace design, as well as examine if a completely integrated platform can provide increased efficiencies, trust and a better overall user experience for consumers within the residential rental industry.

Project Aim and Objectives (300 words)

Target of this Project:

Create an effective, safe and secure system for helping landlords rent out property directly to tenants, rather than having to use real estate agent or property management company, thus making landlord/tenant business more efficient, reduce the risk to parties involved, enhance trustworthiness, provide better service to landlords and tenants throughout process of renting residential property.

Purpose of this Project: Identify common problems encountered with traditional methods for renting properties through 3rd party properties rental platforms and real estate agents; develop a literature study on digital disintermediation, peer-to-peer rental market systems, 2-sided (two-way) service platform model, internet property rental transaction trust systems; create a scalable architecture to support various aspects of the rental process including the creation/management of property listings, communication, scheduling of viewings, payment and analytics reporting; design and build a website that provides all of these features in an easy to use manner; build integrated credit card processing capabilities with Stripe Connect; develop landlord analytics dashboard(s) and landlord verification system that provides improved trust and transparency between landlords and tenants.

To conduct functional, integration, usability, and performance testing to evaluate the effectiveness of the proposed solution. To assess whether the platform can reduce administrative complexity and support more efficient landlord-to-tenant interactions compared with traditional approaches.

Methodology/approach (400 words)

The DSR research method will be used for the project because its purpose is designing, developing and evaluating a technological artefact which solves an actual problem. DSR is very appropriate for computing science projects, as it provides a structured method of identifying the problem, designing the solution, implementing the solution and evaluating it.

Agile Scrum will be used as the methodology for developing the software of the project because it enables iterative development, provides for continuous feedback, and delivers functionality in incremental releases. The project will be broken down into six development sprints, where each sprint will focus on the development of a major subsystem of DirectLet. Each sprint will be completed with testing and review activities to track progress and identify areas for improvement before proceeding to the next sprint.

The Model View Controller (MVC) architecture will be used as a guideline in constructing DirectLet's software. The application's data and the interaction with the MongoDB database will be handled by the Model layer of architecture, while the View layer of architecture will be created using React to provide a user interface for each of the three types of users (i.e. landlords, tenants, and administrators). The Controller layer of architecture will implement the application's business logic and all of the GraphQL resolver functions that process requests and coordinate data exchange between the user interface and database. The MVC architecture creates a separation of concerns resulting in improved maintainability, scalability and testability.

The React front end will implement a Component-Based Architecture. The user interface elements (i.e. property cards, search filters, messaging windows, dashboards, navigation components) will be developed as reusable components, this provides for improved code reuse, easier maintenance, and future system enhancements.

The back end will use the Repository Pattern, meaning that all database operations will be separated from the business logic (this provides for increased flexibility since the capability to access data can be changed without affecting the services or controllers available in the application). The repository pattern also provides for more cleanly organized code and easier unit testing.

Construction will take place over a total of this phases,

Phase 1 — User Authentication, Role-Based Access Control, MongoDB Schema Design

Phase 2 — Property Listing Management, Image Uploads, Advanced Property Search, Map Integration.

Phase 3 — Real Time Messaging and Property Viewing Scheduling

Phase 4 — Integration of Stripe Connect to facilitate Rent Collection and Deposit Payments

Phase 5 — Development of Landlord Analytics Dashboard

Phase 6 — Introduction of Verification Mechanisms and Thorough Testing

The platform will be built using React, Node.Js, GraphQL, MongoDB and Stripe Connect. Synthetic datasets for properties will be developed for testing using Faker.js. OpenStreetMap or Google Maps API will be used to provide a geographic component to the application. The evaluation of the application will be performed using functional, integration/usability/performance testing completed with metrics such as response time, task completion rate, search efficiency and user satisfaction.

Legal/ Social/ Ethical Issues (200 words)

The ACM's Code of Ethics and the BCS's Code of Conduct will guide the implementation of the project. Ethical considerations such as privacy, security, equitable treatment and managing accountability for data will be at the forefront of the platform's design and use.

To adhere to legislation such as the UK General Data Protection Regulation (GDPR) and Data Protection Act (2018), the development process will include consideration for the protection of user information processed by the platform, including contact information, property information, and payment data. Security will be considered through technology such as encrypted communication, password-enabled security, and role-based access controls, resulting in appropriate, secure, user-generated privacy features for users.

The possibility of conducting usability testing on volunteer study participants will be pursued. Therefore, an ethics application to the University will be sought prior to the commencement of any evaluation with potential users. Users will be provided with clearly defined information about the study and informed consent will be obtained prior to any participation.

The platform aims to improve both affordability and accessibility within the rental market by decreasing the need for users to pay expensive intermediary fees. However, measures also need to be in place to decrease fraud and protect the lives of vulnerable users. Therefore, verification processes and reporting mechanisms will be part of the overall system design.

Looking at the overall project from a margin, ethical, legal, social, and professional will ensure that the project is developed in a socially responsible manner.

Project & Risk Management (300 words)

It is very important to manage the project effectively in order to complete it successfully within the required timeline. Risks will continue to be monitored during project development, and mitigation plans will be created where they are needed.

Health and Safety Risks
This project will be primarily made up of standard software development activities with standard computer hardware, so the potential for health and safety risks from this project are minimal and mostly associated with long periods of using computers. Appropriate workstation ergonomics and break periods will be regularly employed to minimise these risk factors.

Management Risks
The main management risk is that the new development and testing tasks will not be completed in a timely manner on the project schedule. The project manager will mitigate this risk by completing the development and testing tasks associated with each of the project's clearly defined phases, using milestone checks and periodic progress assessments to track progress. By focusing on completing the core functionality, the project will be able to deliver a working prototype, even if the timeline to complete the project will result in delays executing the project advanced functionality.

Technical Risk
The project team identified a number of technical risks. Integrating with third party services, such as Stripe Connect and mapping APIs could introduce compatibility issues. To mitigate these risks the project team will create early prototypes of the third party services and test them out.

As an application grows in complexity, there can be performance issues; to reduce the potential for this, performance testing and optimization will be completed at all points during the development life cycle.

Design errors in the database could cause scaleability problems and future maintenance problems. To help mitigate the risk associated with this, close attention will be paid to schema design, the execution of validation tests, and obtaining feedback from supervisors.

Using only the generated datasets might create a level of unrealism in evaluating the product. To help offset the effect of that, synthetic datasets will be developed to replicate what would occur in the actual rental environment.
Being proactive in identifying/managing these risks will greatly increase the chance that a solution will be produced that is reliable, functional, and academically sound.

References in IEEE format (Approx 5 References)

[1] S. R. Jones and M. K. Turner, “Digital Disintermediation in the UK Property Market,” in Proceedings of the IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC), 2023, pp. 1–8.

[2] L. Chen, H. Wang, and J. Zhang, “Optimizing Two-Sided Platform Design for Real Estate Marketplaces,” IEEE Transactions on Engineering Management, vol. 71, pp. 4521–4535, 2024.

[3] M. A. Rahman, A. Islam, and S. Karim, “Blockchain-Based Property Rental Framework with Smart Contracts,” in Proceedings of the IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 2023, pp. 1–5.

[4] T. Yamamoto and K. Suzuki, “Critical Success Factors for Peer-to-Peer Property Rental Platforms,” in Proceedings of the IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), 2024, pp. 1120–1125.

[5] G. Rossi, F. Cabot, and J. Pereira, “Design Science Research in Information Systems and Software Engineering,” ACM Computing Surveys, vol. 56, no. 4, pp. 1–30, 2024.